Assignment 5

Name - Neelay Shah
PRN - 123B1B261

1. Problem Statement

To design and develop a web-based form using React.js that collects user details through multiple input fields and saves the submitted data into a MongoDB database using a backend built with Node.js and Express.js.

2. Aim

To build a full-stack web application using the MERN stack that captures user input from a React form and stores it in a MongoDB database.

3. Objectives

1. To understand how the MERN stack works as a complete architecture.
2. To design an interactive form in React with multiple input fields.
3. To create a backend server using Node.js and Express.js for handling requests.
4. To store user-submitted data in a MongoDB database.
5. To connect the frontend and backend using RESTful APIs.

4. Theory

MERN Stack

The MERN stack is a widely used JavaScript-based framework for developing modern web applications. It includes:

- **MongoDB**
  A NoSQL database that stores data in a flexible, JSON-like format called documents.
- **Express.js**
  A lightweight framework for Node.js that simplifies building APIs and handling server-side logic.
- **React.js**
  A library used to build dynamic and interactive user interfaces, especially for single-page applications.
- **Node.js**
  A runtime environment that allows JavaScript to run on the server side.

Application Workflow

The working of the application follows this flow:

React Form (Frontend)
↓
Axios HTTP Request
↓
Express.js API (Backend)
↓
MongoDB Database

Here, the frontend sends user data using Axios to the backend API, which processes and stores it in the database.

5. System Requirements

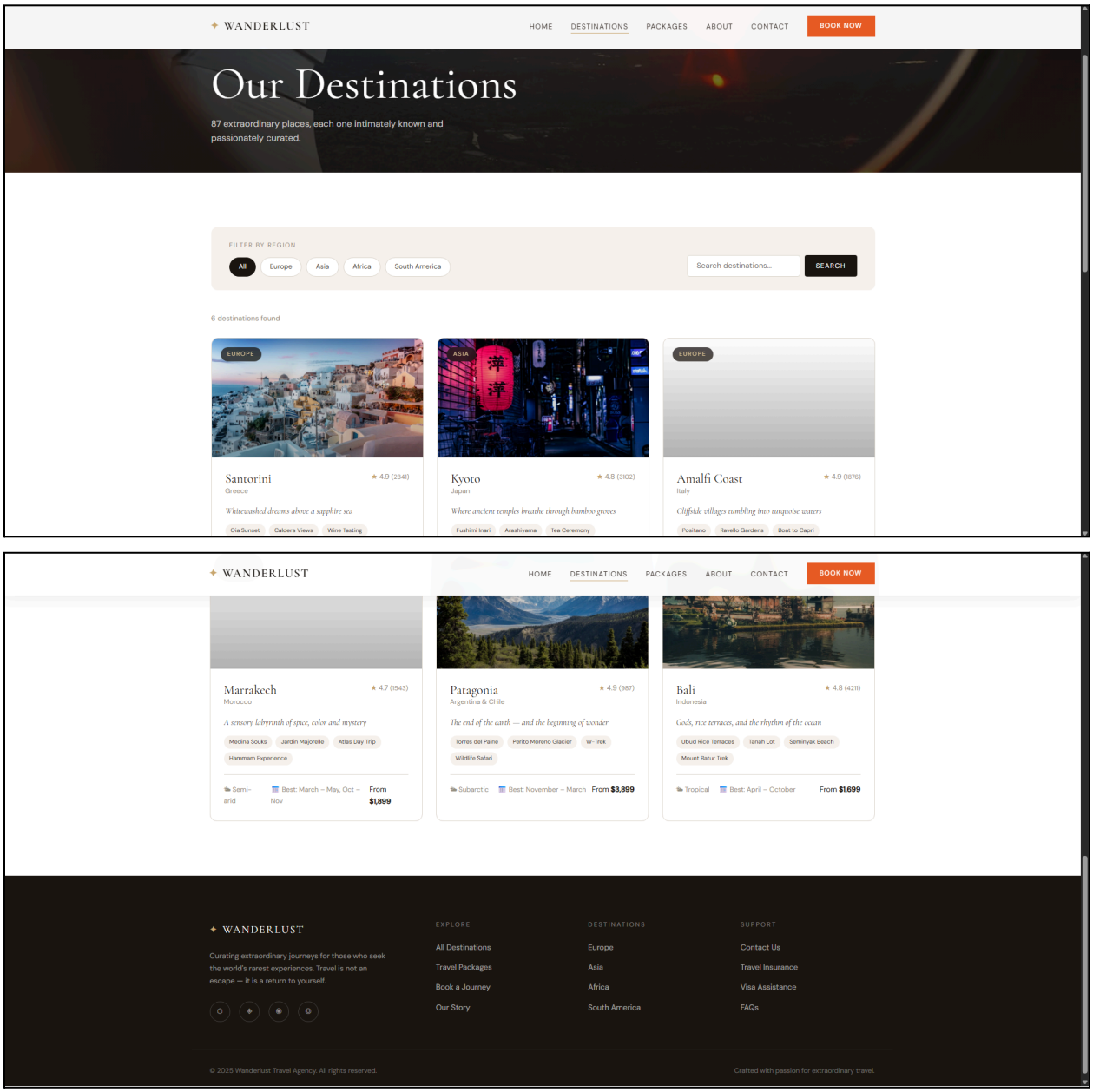
Software Requirements

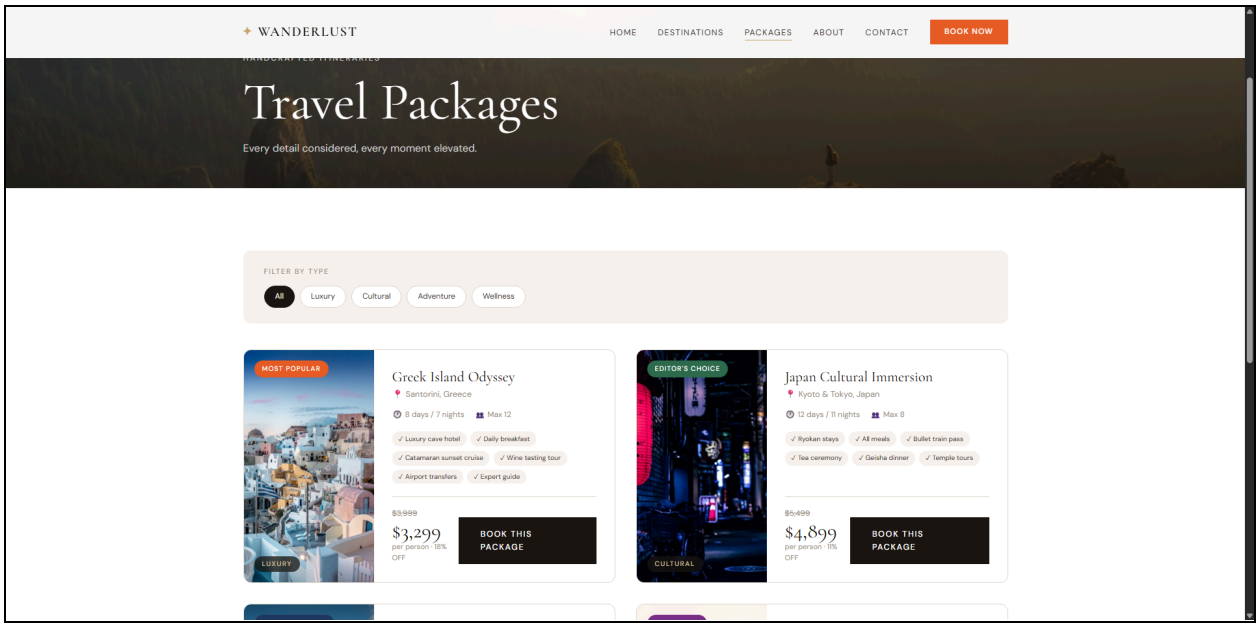
- Node.js
- MongoDB
- React.js
- Express.js
- Axios
- VS Code or any code editor

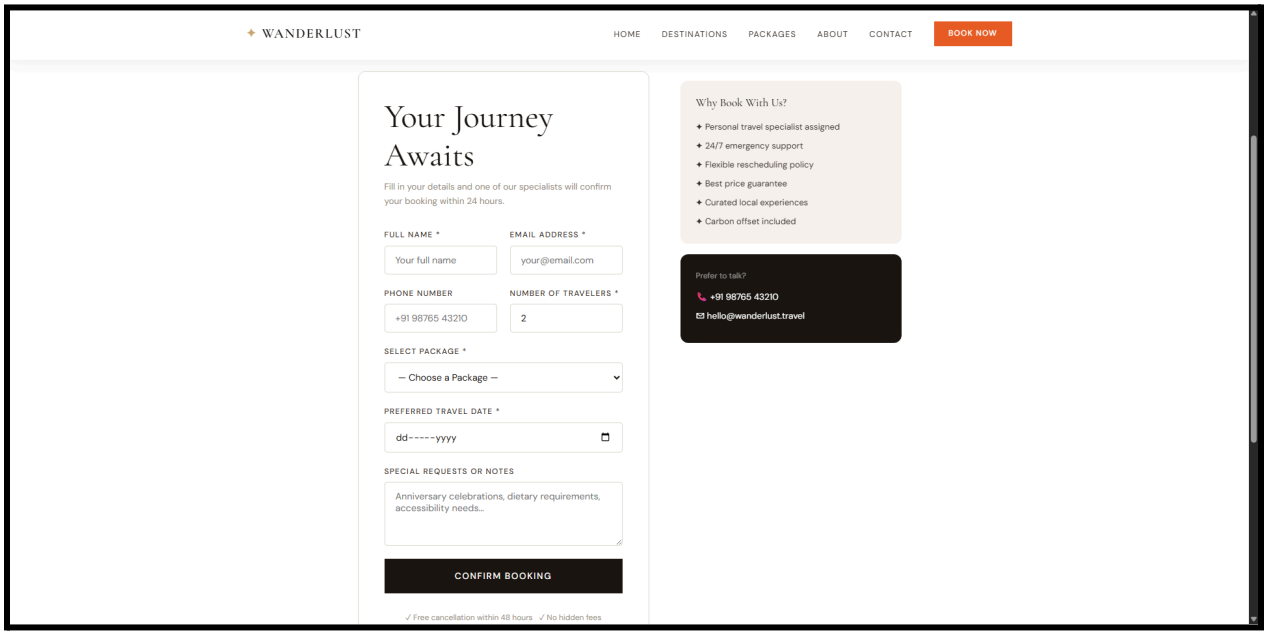
Hardware Requirements

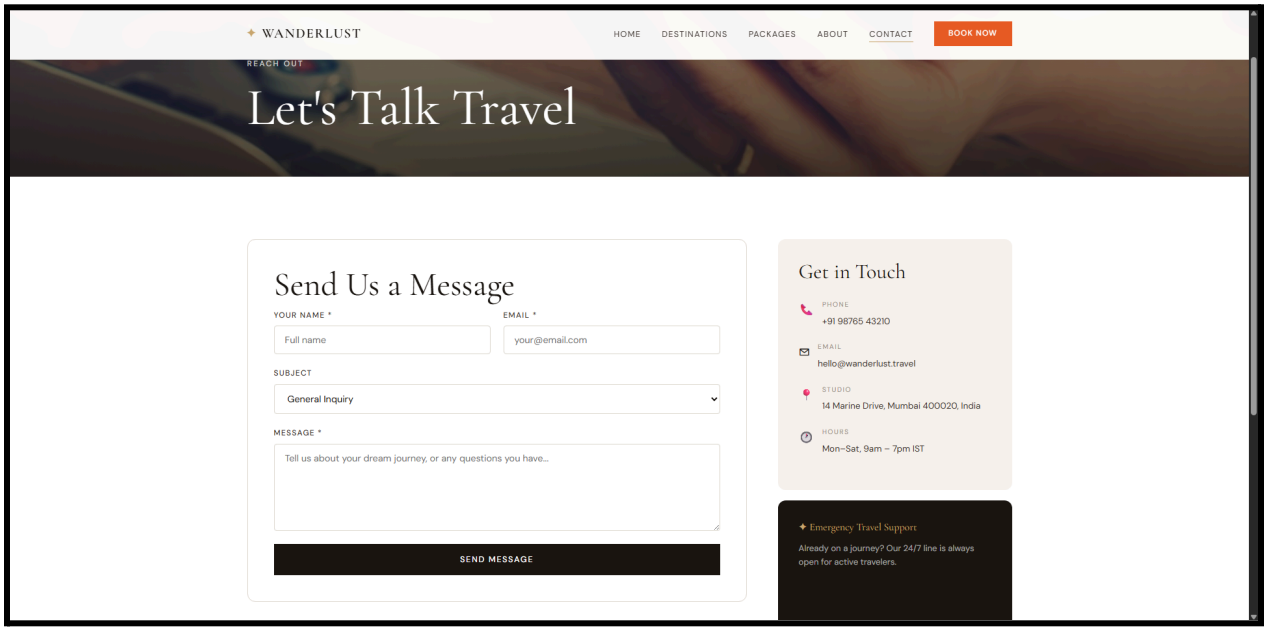
- Minimum 4 GB RAM
- Operating System: Ubuntu, Windows, or macOS
- Internet connection

Output


WANDERLUST
HOME
DESTINATIONS
PACKAGES
ABOUT
CONTACT
BOOK NOW
Our Destinations
87 extraordinary places, each one intimately known and passionately curated.
FILTER BY REGION
All
Europe
Asia
Africa
South America
Search destinations...
SEARCH
6 destinations found
EUROPE
Santorini
Greece
4.9 (2341)
Whitewashed dreams above a sapphire sea
Cia Sunset
Caldera Views
Wine Tasting
ASIA
Kyoto
Japan
4.8 (3102)
Where ancient temples breathe through bamboo groves
Fushimi Inari
Arashiyama
Tea Ceremony
EUROPE
Amalfi Coast
Italy
4.9 (1876)
Cliffside villages tumbling into turquoise waters
Positano
Ravello Gardens
Boat to Capri
WANDERLUST
HOME
DESTINATIONS
PACKAGES
ABOUT
CONTACT
BOOK NOW
Marrakech
Morocco
4.7 (1543)
A sensory labyrinth of spice, color and mystery
Medina Souks
Jardin Majorelle
Atlas Day Trip
Hammam Experience
Semi-arid
Best: March – May, Oct – Nov
From $1,899
Patagonia
Argentina & Chile
4.9 (987)
The end of the earth — and the beginning of wonder
Torres del Paine
Perito Moreno Glacier
W-Trek
Wildlife Safari
Subarctic
Best: November – March
From $3,899
Bali
Indonesia
4.8 (4211)
Gods, rice terraces, and the rhythm of the ocean
Ubud Rice Terraces
Tanah Lot
Seminyak Beach
Mount Batur Trek
Tropical
Best: April – October
From $1,699
WANDERLUST
Curating extraordinary journeys for those who seek the world's rarest experiences. Travel is not an escape — it is a return to yourself.
EXPLORE
All Destinations
Travel Packages
Book a Journey
Our Story
DESTINATIONS
Europe
Asia
Africa
South America
SUPPORT
Contact Us
Travel Insurance
Visa Assistance
FAQs
© 2025 Wanderlust Travel Agency. All rights reserved.
Crafted with passion for extraordinary travel.


WANDERLUST
HOME
DESTINATIONS
PACKAGES
ABOUT
CONTACT
BOOK NOW
Travel Packages
Every detail considered, every moment elevated.
FILTER BY TYPE
All
Luxury
Cultural
Adventure
Wellness
MOST POPULAR
Greek Island Odyssey
Santorini, Greece
8 days / 7 nights
Max 12
✓ Luxury cave hotel
✓ Daily breakfast
✓ Catamaran sunset cruise
✓ Wine tasting tour
✓ Airport transfers
✓ Expert guide
$3,999
$3,299
per person · 18% OFF
BOOK THIS PACKAGE
LUXURY
EDITOR'S CHOICE
Japan Cultural Immersion
Kyoto & Tokyo, Japan
12 days / 11 nights
Max 8
✓ Ryokan stays
✓ All meals
✓ Bullet train pass
✓ Tea ceremony
✓ Geisha dinner
✓ Temple tours
$5,499
$4,899
per person · 11% OFF
BOOK THIS PACKAGE
CULTURAL


WANDERLUST
HOME
DESTINATIONS
PACKAGES
ABOUT
CONTACT
BOOK NOW
Your Journey Awaits
Fill in your details and one of our specialists will confirm your booking within 24 hours.
FULL NAME *
Your full name
EMAIL ADDRESS *
your@email.com
PHONE NUMBER
+91 98765 43210
NUMBER OF TRAVELERS *
2
SELECT PACKAGE *
— Choose a Package —
PREFERRED TRAVEL DATE *
dd-----yyyy
SPECIAL REQUESTS OR NOTES
Anniversary celebrations, dietary requirements, accessibility needs...
CONFIRM BOOKING
✓ Free cancellation within 48 hours ✓ No hidden fees
Why Book With Us?
✦ Personal travel specialist assigned
✦ 24/7 emergency support
✦ Flexible rescheduling policy
✦ Best price guarantee
✦ Curated local experiences
✦ Carbon offset included
Prefer to talk?
+91 98765 43210
hello@wanderlust.travel


WANDERLUST
HOME
DESTINATIONS
PACKAGES
ABOUT
CONTACT
BOOK NOW
REACH OUT
Let's Talk Travel
Send Us a Message
YOUR NAME *
Full name
EMAIL *
your@email.com
SUBJECT
General Inquiry
MESSAGE *
Tell us about your dream journey, or any questions you have...
SEND MESSAGE
Get in Touch
PHONE
+91 98765 43210
EMAIL
hello@wanderlust.travel
STUDIO
14 Marine Drive, Mumbai 400020, India
HOURS
Mon–Sat, 9am – 7pm IST
✦ Emergency Travel Support
Already on a journey? Our 24/7 line is always open for active travelers.

7. Result

The React form successfully collects user information including:

- Name
- Email
- Age
- Phone
- Address
- Gender
- Department

After submitting the form, the data is sent to the Node.js backend server and stored in the MongoDB database.

8. Conclusion

The experiment successfully demonstrates how to build a full-stack web application using the MERN stack. The project shows how React forms interact with a backend API and how data is stored in MongoDB. This implementation helps understand modern web development practices and REST API communication.